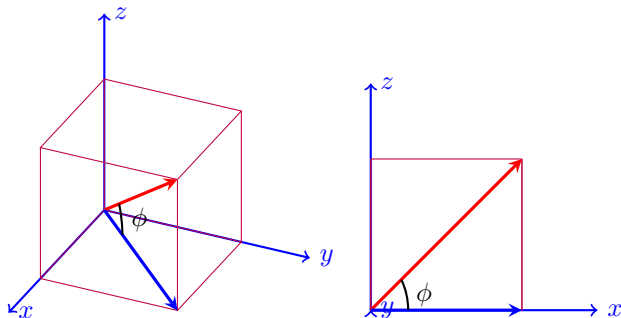


1. $\mathbf{x} \cdot \mathbf{y} = (x_1, x_2, \dots, x_n) \cdot (y_1, y_2, \dots, y_n) = x_1 y_1 + x_2 y_2 + \dots + x_n y_n$ and $\mathbf{x} \cdot \mathbf{y} = \|\mathbf{x}\| \|\mathbf{y}\| \cos \theta$.
- b. $(2, 1) \cdot (-1, 1) = 2(-1) + 1(1) = -2 + 1 = -1$. $\|(2, 1)\| = \sqrt{2^2 + 1^2} = \sqrt{5}$ and $\|(-1, 1)\| = \sqrt{(-1)^2 + 1^2} = \sqrt{2}$. So then $\theta = \cos^{-1}(-1/\sqrt{10})$ ($\approx 108.435^\circ$).
- c. $(1, 8) \cdot (7, 4) = -25$. $\theta = \cos^{-1}(-25/\|(1, 8)\| \|(7, 4)\|) = \cos^{-1}(-25/\sqrt{65}\sqrt{65}) = \cos^{-1}(-5/13)$ ($\approx 112.62^\circ$).
- d. $(1, 4, -3) \cdot (5, 1, 3) = 1(5) + 4(1) + (-3)(3) = 5 + 4 - 9 = 0$. So the vectors are orthogonal, i.e., $\theta = \pi/2 = 90^\circ$.
2. $\text{proj}_{\mathbf{y}} \mathbf{x} = \frac{\mathbf{x} \cdot \mathbf{y}}{\|\mathbf{y}\|^2} \mathbf{y}$ and $\text{proj}_{\mathbf{x}} \mathbf{y} = \frac{\mathbf{x} \cdot \mathbf{y}}{\|\mathbf{x}\|^2} \mathbf{x}$.
- b. Using results from above,
 $\text{proj}_{\mathbf{y}} \mathbf{x} = \frac{-1}{(\sqrt{2})^2} (-1, 1) = -\frac{1}{2}(-1, 1)$ and $\text{proj}_{\mathbf{x}} \mathbf{y} = \frac{-1}{(\sqrt{5})^2} (2, 1) = -\frac{1}{5}(2, 1)$.
- d. The vectors are orthogonal, so $\text{proj}_{\mathbf{y}} \mathbf{x} = \text{proj}_{\mathbf{x}} \mathbf{y} = 0$.
4. Note that scaling and position does not affect the problem. So without loss of generality we can assume that vectors $(0, 0, 0)$, $(1, 0, 0)$, $(0, 1, 0)$, $(1, 1, 0)$, $(0, 0, 1)$, $(1, 0, 1)$, $(0, 1, 1)$, and $(1, 1, 1)$ represent the eight vertices of the cube. A long diagonal and a face angle are given by $(1, 1, 1)$ and $(1, 0, 0)$ respectively. $(1, 1, 1) \cdot (1, 1, 0) = 2$, $\|(1, 1, 1)\| = \sqrt{3}$, and $\|(1, 1, 0)\| = \sqrt{2}$. This gives $\cos^{-1}(2/\sqrt{6}) \approx 35.3^\circ$.



7. The vectors $2\mathbf{x} + 3\mathbf{y}$ and $\mathbf{x} - \mathbf{y}$ are orthogonal if and only if their dot product is 0. The dot product is distributive and $\mathbf{x} \cdot \mathbf{x} = \|\mathbf{x}\|^2$ so then

$$\begin{aligned}
 (2\mathbf{x} + 3\mathbf{y}) \cdot (\mathbf{x} - \mathbf{y}) &= 2\mathbf{x} \cdot \mathbf{x} - 2\mathbf{x} \cdot \mathbf{y} + 3\mathbf{y} \cdot \mathbf{x} - 3\mathbf{y} \cdot \mathbf{y} \\
 &= 2\|\mathbf{x}\|^2 + \mathbf{y} \cdot \mathbf{x} - 3\|\mathbf{y}\|^2 \\
 &= 2(2) + \mathbf{y} \cdot \mathbf{x} - 3.
 \end{aligned}$$

The angle between $\mathbf{x}$ and $\mathbf{y}$ is $3\pi/4 = \cos^{-1}(\mathbf{x} \cdot \mathbf{y} / \sqrt{2}) \Rightarrow \mathbf{x} \cdot \mathbf{y} = -1$ (solve for $\mathbf{x} \cdot \mathbf{y}$). Thus

$$2(2) + \mathbf{y} \cdot \mathbf{x} - 3 = 4 - 1 - 3 = 0$$

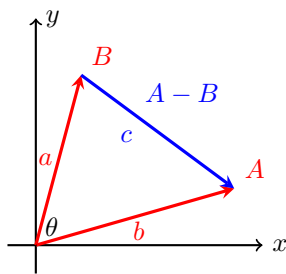
Which tells us that $2\mathbf{x} + 3\mathbf{y}$ and $\mathbf{x} - \mathbf{y}$ are orthogonal.

11. $\mathbf{x}$ is orthogonal to each $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$ implies that $\mathbf{x} \cdot \mathbf{v}_i = 0$ for each $i = 1, 2, \dots, k$. So then by the distributive property of the dot product,

$$\begin{aligned}
 \mathbf{x} \cdot (c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \dots + c_k \mathbf{v}_k) &= c_1 \mathbf{x} \cdot \mathbf{v}_1 + c_2 \mathbf{x} \cdot \mathbf{v}_2 + \dots + c_k \mathbf{x} \cdot \mathbf{v}_k \\
 &= c_1(0) + c_2(0) + \dots + c_k(0) \\
 &= 0 + 0 + \dots + 0 = 0.
 \end{aligned}$$

So then $\mathbf{x}$ is orthogonal to $c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \dots + c_k \mathbf{v}_k$ for any $c_1, c_2, \dots, c_k \in \mathbb{R}$.

14 Let A, B be vectors as illustrated below.



$b = \|A\|$, $a = \|B\|$, $c = \|A - B\|$, and $\|A\|\|B\|\cos\theta = A \cdot B$. So then

$$\begin{aligned}
 a^2 + b^2 - 2ab(\cos\theta) &= \|B\|^2 + \|A\|^2 - 2\|B\|\|A\|\left(\frac{A \cdot B}{\|A\|\|B\|}\right) \\
 &= \|B\|^2 + \|A\|^2 - 2A \cdot B \\
 &= B \cdot B + A \cdot A - 2A \cdot B \\
 &= (B - A)^2 = (A - B)^2 = \|A - B\|^2 \\
 &= c^2
 \end{aligned}$$

(c is the length between A and B so then $c = \|A - B\|$).

18. The Cauchy-Schwarz inequality says that $|\mathbf{x} \cdot \mathbf{y}| \leq \|\mathbf{x}\|\|\mathbf{y}\|$. Following the hint we use

$$\begin{aligned}
 \|\mathbf{x} + \mathbf{y}\|^2 &= (\mathbf{x} + \mathbf{y}) \cdot (\mathbf{x} + \mathbf{y}) = \|\mathbf{x}\|^2 + 2\mathbf{x} \cdot \mathbf{y} + \|\mathbf{y}\|^2 \\
 &\leq \|\mathbf{x}\|^2 + 2|\mathbf{x} \cdot \mathbf{y}| + \|\mathbf{y}\|^2 \\
 &\leq \|\mathbf{x}\|^2 + 2\|\mathbf{x}\|\|\mathbf{y}\| + \|\mathbf{y}\|^2 \text{ (by Cauchy-Schwarz)} \\
 &\leq (\|\mathbf{x}\| + \|\mathbf{y}\|)^2 \text{ (factor)}
 \end{aligned}$$